Fiscal Year:	FY 2010	**Task Last Updated:**	FY 09/12/2013
PI Name:	Platts, Steven H. Ph.D.		
Project Title:	Hypovolemia as a Model of Space Flight: Cardiovascular Effects and Countermeasures		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Johnson Space Center		
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PI Address 2:	Biomedical Research and Environmental Sciences Division		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2006	**End Date:**	09/29/2010
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
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Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Orthostatic intolerance, an impaired ability to appropriately maintain blood pressure in the upright posture, is a consistent finding among astronauts following both short- and long-duration space flight. Orthostatic intolerance is considered to be a health and safety risk during re-entry, landing, and post-flight activities, and therefore has been subject of numerous studies to elucidate its etiology and to develop and test countermeasures. Plasma volume loss, which occurs early in after space flight and remains reduced on landing day, is believed to contribute to orthostatic intolerance and other measures of space flight-induced cardiovascular deconditioning, including cardiac atrophy. The objectives of this study were to determine whether a pharmacologically-induced plasma volume loss would increase the rate of presyncope during a 30-minute 80° head-up tilt test in normal healthy subjects and result in apparent cardiac atrophy measured with 3-d echocardiography compared to the normovolemic condition. Originally, this project also		

	sought to determine the effectiveness of pharmacological countermeasures against orthostatic intolerance, but those arms of the study were canceled prior to the start of data collection.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The objectives of this study were to determine whether a pharmacologically-induced plasma volume loss would increase the rate of presyncope during a 30-minute 80° head-up tilt test in normal healthy subjects and result in apparent cardiac atrophy measured with 3-d echocardiography compared to the normovolemic condition. Originally, this project also sought to determine the effectiveness of pharmacological countermeasures against orthostatic intolerance, but those arms of the study were canceled prior to the start of data collection. The infusion of Lasix (furosemide) induced a plasma volume loss which was comparable to that observed in astronauts following space flight. However, despite a mildly increased sympathetic response (catecholamine levels) and smaller stroke volumes during tilt, the rate of presyncope during the tilt test in this sample of normal healthy subjects did not increase when the subjects were hypovolemic. Additionally, while some indices of cardiac function were reduced, there was no measureable cardiac atrophy observed with hypovolemia. Therefore, although pharmacologically-induced plasma volume loss produces some changes in cardiovascular function which are similar to space flight, by itself plasma volume reductions do not fully explain space flight-induced orthostatic intolerance or cardiac atrophy. Plasma volume loss contributes to, but does not fully explain, space flight-induced orthostatic intolerance and decreased cardiac function. The hypovolemia model may be used as a screening tool for some countermeasures to orthostatic intolerance, but success of a countermeasure in this model does not guarantee success after actual space flight. When cost-effective, other space flight analogs (i.e. bed rest) may provide more confidence in the results of countermeasure testing. [Ed. note 9/12/13: information from PI's Task Book Initial Entry Form, provided September 2013]
Bibliography Type:	Description: (Last Updated: 03/01/2018)